



TMS

TRANSCRANIAL
MAGNETIC STIMULATION

THE TRUE SPECIALIST IN TMS THERAPY



REMED
Rehabilitation Medical Company

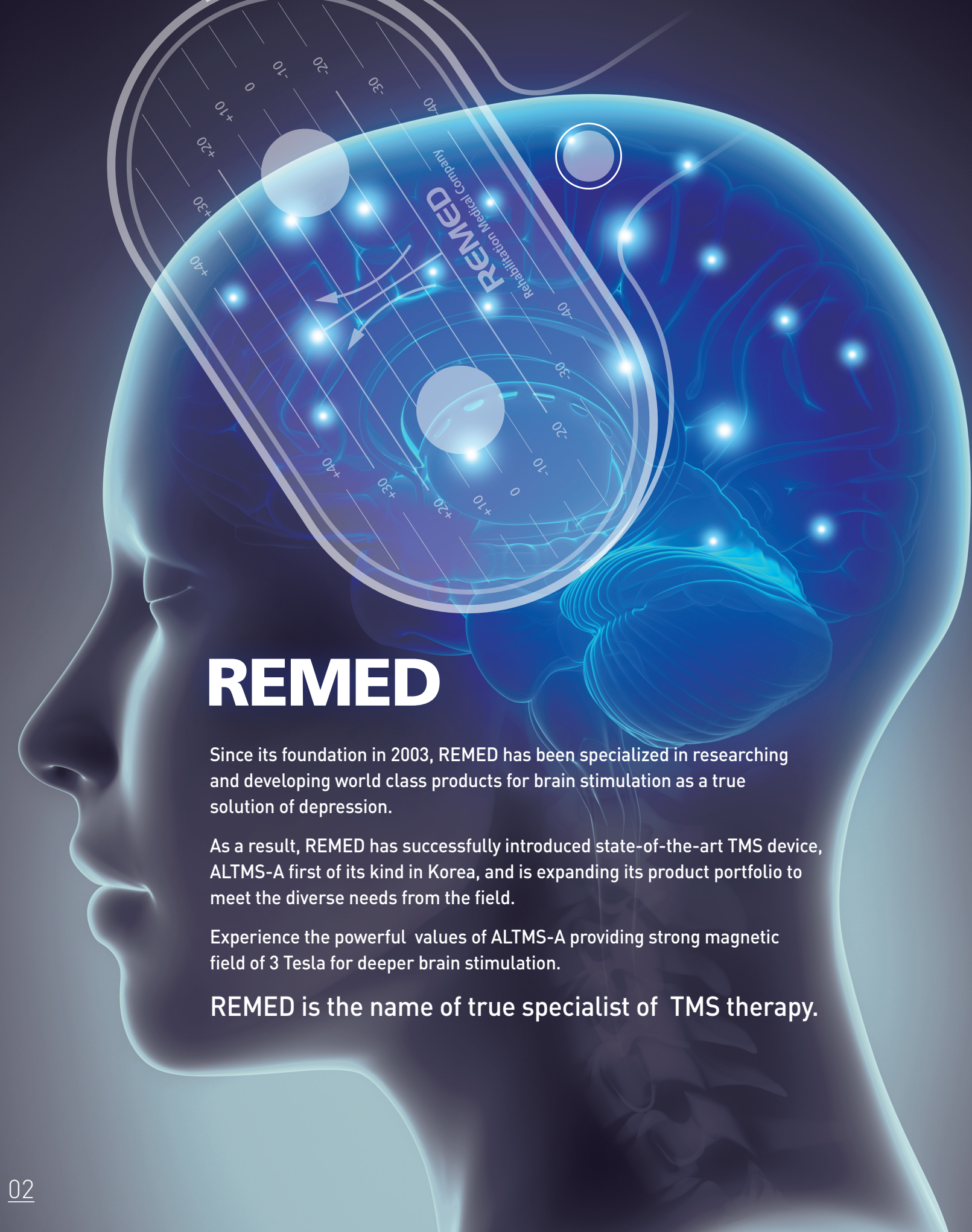
WHEN YOUR PATIENTS ARE SUFFERING FROM



**REMED PROVIDES TRUE
SOLUTION FOR THEM AS GENUINE
SPECIALIST OF TMS THERAPY.**

REMED

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REMED

Since its foundation in 2003, REMED has been specialized in researching and developing world class products for brain stimulation as a true solution of depression.

As a result, REMED has successfully introduced state-of-the-art TMS device, ALTMS-A first of its kind in Korea, and is expanding its product portfolio to meet the diverse needs from the field.

Experience the powerful values of ALTMS-A providing strong magnetic field of 3 Tesla for deeper brain stimulation.

REMED is the name of true specialist of TMS therapy.

WHY TMS THERAPY?

(TRANSCRANIAL MAGNETIC STIMULATION)

Transcranial Magnetic Stimulation(TMS) is a safe, non-invasive therapy method that uses rapidly changing magnetic field to stimulate the neurons in the brain. It is indicated for the treatment of Major Depressive Disorder(MDD), and other certain conditions, for adult patients who have failed to achieve satisfactory improvement from prior antidepressant medication.

Also, TMS is widely accepted as a unique research tool for the investigation of broad variety of issues in different fields:

Cognitive Neuroscience- in studying mental processes related to memory, speech, perception and functional connection

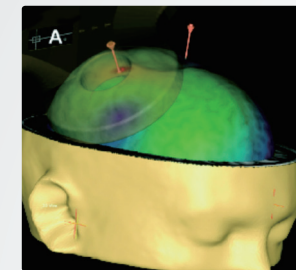
Psychiatry- to influence specific brain function within the dorsolateral prefrontal cortex

Neurophysiology- used in the stimulation of the peripheral and central nerve pathways

Rehabilitation- used in the promotion of muscle recovery and the relief of nerve spasticity

TECHNICALLY PROVEN MECHANISM OF TMS

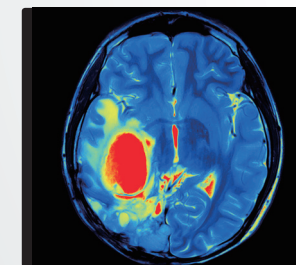
TMS therapy is an outpatient procedure and does not require anesthesia or sedation, so the patient can go back to daily life right after treatment. Unlike other treatments, such as electro-convulsive therapy(ECT), TMS does not cause cognitive side effects or memory problems.



Time-varying electrical current in a coil is produced.



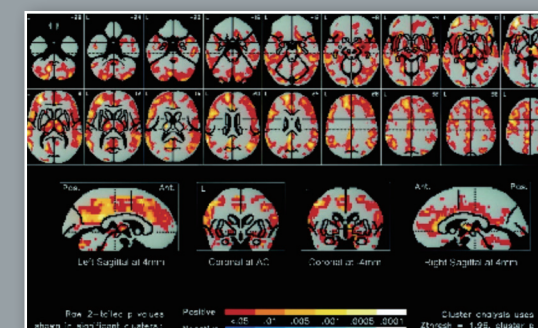
More focused magnetic field from figure of 8 shaped transducer passes unimpeded through skull.



Neurons in the cortex of brain (PFC) is stimulated to lead to behavioral changes.

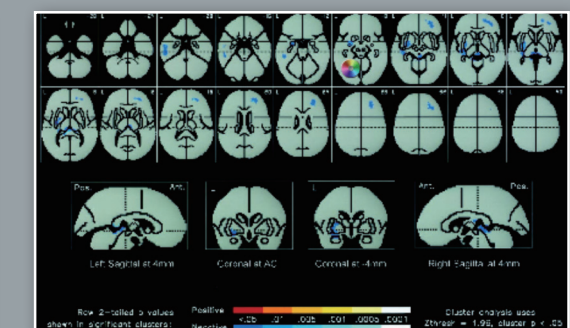
MRI images for patients with Depression and ADHD after TMS treatment

20 Hz



Patient with Depression

1 Hz



Patient with ADHD

The true specialist of TMS for deeper brain stimulation

ALTMS[®]

Most reliable & powerful TMS Device

Experience the powerful values of ALTMS[®] providing strong magnetic field of 3 Tesla for deeper brain stimulation.



Features & Benefits

- Convenient & easy operation by touch screen color display
- Unique design of 8 shaped transducer for focused brain stimulation
- Patented oil circulating transducer for intensive & non-tiring treatment
- 2 stimulation modes for diverse applications: Single, Repetitive

Technical Specifications

Peak Magnetic Energy	1 Tesla peak @ 100% output
Frequency	Maximum 30Hz Minimum 0.1Hz
Train Duration	0.1–1Hz: 1–1800 sec, 2–30Hz: 1–20 sec increment
Output waveform	Symmetric biphasic pulse
Weight	61Kg
Power Supply	120V, 60Hz
Software	Integrated color touch screen display 2 Stimulation mode (Single, Repetitive)

For more detailed clinical indications, please refer to the following reference site:
<https://www.ncbi.nlm.nih.gov/pubmed/>



ALTMS[®]



“ Side effects for patients with drug therapy “

Insomnia, Fatigue Nausea, Weight Gain, Sexual Dysfunction

“ Let your patients smile again with TMS therapy “



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Original Article
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The Efficacy of Miniaturized Repetitive Transcranial Magnetic Stimulation in Patients with Depression

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Objective: The present study aimed to evaluate the efficacy of repetitive transcranial magnetic stimulation (rTMS) at a high frequency using a miniaturized device compared to standard rTMS and sham rTMS for the treatment of depression.

Methods: Thirty-four patients with depression were randomly assigned to either 15 days of miniaturized, standard, or sham rTMS. The stimulation consisted of 60 trains of 3 seconds at 10 Hz for 30 minutes. Clinical measures were assessed at baseline and on the final day of the stimulation.

Results: A post-hoc analysis of patients showed a significant main effect of time and a time by group interaction on Hamilton Rating Scale for Depression scores. There were no significant correlations between individual clinical disorders and changes of clinical outcomes. Our results revealed a significant reduction in the Hamilton Rating Scale for Depression in the miniaturized and standard group compared to the sham group.

Conclusion: The antidepressant effect of miniaturized rTMS using subthreshold stimulation was comparable to that of standard stimulation.

KEY WORDS: Depression; Transcranial magnetic stimulation; Antidepressants.

INTRODUCTION

Providing both existing treatment options and greater access to treatment may be more clinically significant than the development of entirely new treatment modalities, because many individuals who have depression do not respond to their symptoms. Many alternative treatments for patients with major depressive disorder (MDD) have emerged in recent years. Among these, repetitive transcranial magnetic stimulation (rTMS), a non-invasive, non-pharmacological technique, has been demonstrated to be safe and effective for the treatment of MDD. Previous studies have demonstrated the antidepressant effect of high-frequency rTMS when applied to the left dorsolateral prefrontal cortex (DLPFC) specifically.¹⁻³ Many outpatients being treated for depression fail to complete their recommended course of treatment.⁴ Because the penetration of magnetic current is greater in some cases, e.g., increased major depressive episodes, improving the continuity of treatment for these patients is highly critical to their care.⁵ Standard rTMS protocols require daily stimulation for several weeks and may consequently lead to elevated treatment discontinuity rates.^{6,7} Deep-seated sites with conventional high-frequency rTMS and sham rTMS are comparable.^{8,9} In daily clinic visits are required for both and may be an obstacle to adequate treatment continuity. The use of miniaturized devices, which are smaller and lighter than standard ones, may thus serve as a good solution for outpatient treatment. Patients with MDD could potentially use these devices at home without regular visits to clinics, lowering rates of disease and thus improving treatment adherence.

Treatment with rTMS is characterized by many variables. It is a non-invasive, non-pharmacological technique that is safe and effective for the treatment of MDD. The use of miniaturized devices, which are smaller and lighter than standard ones, may thus serve as a good solution for outpatient treatment. Patients with MDD could potentially use these devices at home without regular visits to clinics, lowering rates of disease and thus improving treatment adherence.

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Abstract
We conducted a 4-week randomized, sham-controlled, single-blind, parallel-group trial to evaluate the effect of repetitive transcranial magnetic stimulation (rTMS) delivered to the left dorsolateral prefrontal cortex (DLPFC) on functional connectivity and body weight in obese adults with obesity.

Materials and methods: Of the 45 volunteers with obesity, aged between 18 and 70 years, body mass index (BMI) ≥ 25 kg/m² according to the obesity criteria for an Asian population, 36 participants (24 $\pm$ 13.0 years, 894.30 $\pm$ 3.3 kg/m², 77.6% female) completed the 4 weeks of follow-up, including two testing sites (left DLPFC in the real stimulation group and 16 in the sham stimulation group). A total of 18 right sessions of high-frequency rTMS targeting the left DLPFC were provided over a period of 4 weeks (2 sessions per week). Secondary outcomes included, 10 Hz, 1200 pulses (threshold, 2000 pulses over 20 minutes).

Results: Participants in the real stimulation group showed significantly greater weight loss from baseline following the eight sessions of rTMS (-2.53 ± 2.43 kg vs. -0.88 ± 1.23 kg, $p = 0.02$) for real-time brain connectivity measures, the functional connectivity within the right hemisphere network tended to increase with rTMS, and a significant interaction effect was identified for time (pre vs. post) $\times$ rTMS (real vs. sham) in the right hemisphere network ($p = 0.02$, FDR corrected).

Conclusions: We observed that rTMS selectively increased resting-state functional connectivity within the right hemisphere network. Our findings suggest that high-frequency rTMS in the left DLPFC might strengthen the functional network that facilitates food intake and appetite regulation.

KEYWORDS: Obesity; Repetitive transcranial magnetic stimulation; Brain connectivity; Weight loss.

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The effects of repetitive transcranial magnetic stimulation on body weight and food consumption in obese adults: A randomized controlled study

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ABSTRACT

Background: Although some studies have reported significant reductions in food cravings following the single-session of repetitive transcranial magnetic stimulation (rTMS), there is little research on the effects of multi-session of rTMS on food consumption and body weight in obese subjects.

Objective: We conducted a 4-week randomized, sham-controlled, single-blind, parallel-group trial to examine the effect of rTMS on body weight in obese adults.

Methods: Forty-three obese patients (body mass index [BMI] ≥ 25 kg/m²) aged between 18 and 70 years were randomized to the sham or real treatment group (21 in the rTMS group and 22 in the sham treatment group). A total of 8 sessions of rTMS targeting the left dorsolateral prefrontal cortex (DLPFC) was provided over a period of 4 weeks. The primary outcome measure was weight change in kilograms from baseline to 4 weeks. Secondary endpoints included changes in anthropometric measures, cardiovascular risk factors, food intake, and appetite.

Results: Participants in the rTMS group showed significantly greater weight loss from baseline following the 8 sessions of rTMS (-2.75 ± 2.17 kg vs. 0.38 ± 1.0 kg, $p = 0.01$). Consistent with weight loss, there was a significant reduction in fat mass and visceral adipose tissue at week 4 in the rTMS group compared with the control group ($p < 0.01$). After the 8 sessions of rTMS, the rTMS group consumed fewer total kilocalories and carbohydrates per day than the control group ($p < 0.05$).

Conclusions: 8 sessions of 10 Hz rTMS delivered to the left DLPFC were effective in inducing weight loss and decreasing food intake in obese patients.

Trial registration: Clinical trial registered with the Clinical Trials Registry at [http://crd4.org/ct2/115484](#) (NCT03825484).

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Accelerated Transcranial Magnetic Stimulation (TMS) for Depression

Summary

The purpose of this study is to determine if accelerated rTMS treatment over 1.5 days is effective for ameliorating depression in Parkinson's disease.

Description

Objective : The goal of this study is to investigate a new approach to administering repetitive transcranial magnetic stimulation (rTMS) in patients with refractory depression. (Please Note : The original requirement for comorbid Parkinson's disease has been dropped from this study).

Research Plan : This inpatient study will provide an initial test for the hypothesis that accelerated rTMS is an effective treatment for depression. Followup testing will help delineate the time course of response.

Methods : The rTMS treatment site over left dorsolateral prefrontal cortex will be 5.5cm

RTMS may be a good choice for pregnant women with depression

Danqing Zhang · Zeping He

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Abstract Due to the possible adverse effects on fetus, the treatment of pregnant women with depression is a difficult problem in the field of psychiatry. Repetitive transcranial magnetic stimulation (rTMS) may serve as a new choice for pregnant women with depression in the early pregnancy. After systematic treatment, the patient's depressive symptoms significantly relieved, and each of them successfully provided a healthy baby. Up to now, we have not found that rTMS can be harmful to fetus. RTMS may be a good choice for pregnant women with depression.

Keywords: Repetitive transcranial magnetic stimulation (rTMS); Pregnant women; Depression; Treatment.

Dear Sir,

As the symptoms of depression have adverse impacts on pregnant women and fetus, many experts have been looking for a safe and effective treatment for pregnant women with depression.

The United States Food and Drug Administration (FDA) has approved the use of rTMS for the treatment of major depressive disorder (MDD) in non-pregnant women. However, the use of rTMS in pregnant women is still controversial. Some studies have shown that rTMS is safe and effective for pregnant women with depression. However, some studies have shown that rTMS may be harmful to fetus. Therefore, the use of rTMS in pregnant women is still controversial.

RTMS is a new non-pharmacological technique, which is based on the basis of electrical stimulation of the brain. It has been widely used in the treatment of various psychiatric disorders, especially for treating depression. RTMS can enhance the excitability of the brain through adjusting the biological activities of the cerebral cortex and has been widely used in the treatment of various psychiatric disorders, especially for treating depression. RTMS can enhance the excitability of the brain through adjusting the biological activities of the cerebral cortex and has been widely used in the treatment of various psychiatric disorders, especially for treating depression.

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